MEDICAL DATA HISTORY

Prepared By: A Gurukiran

Project Code: PRSQL – 02

Candidate Unique no: CA12202582281

Tools Used: SQL, Power BI.

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1.Project overview

The Medical Data History Analysis project was designed to explore and analyze healthcare data stored in a relational database using SQL. The database consists of multiple interconnected tables containing information about patients, hospital admissions, doctors, and geographical locations. The primary objective of the project was to extract meaningful information from the database and answer a series of business and healthcare-related questions through SQL queries.

The project involved working with real-world database concepts such as data retrieval, filtering, aggregation, joins, grouping, sorting, and data transformation. A total of 34 problem statements were solved, covering both basic and advanced SQL operations. These queries simulated real business scenarios that healthcare organizations may encounter while managing patient records, admission histories, doctor assignments, and demographic information.

Throughout the project, SQL was used to analyze patient demographics, identify admission trends, evaluate diagnosis patterns, examine doctor-patient relationships, and generate statistical summaries from healthcare records. The project also included data cleaning tasks such as handling missing values and formatting data for better readability and reporting.

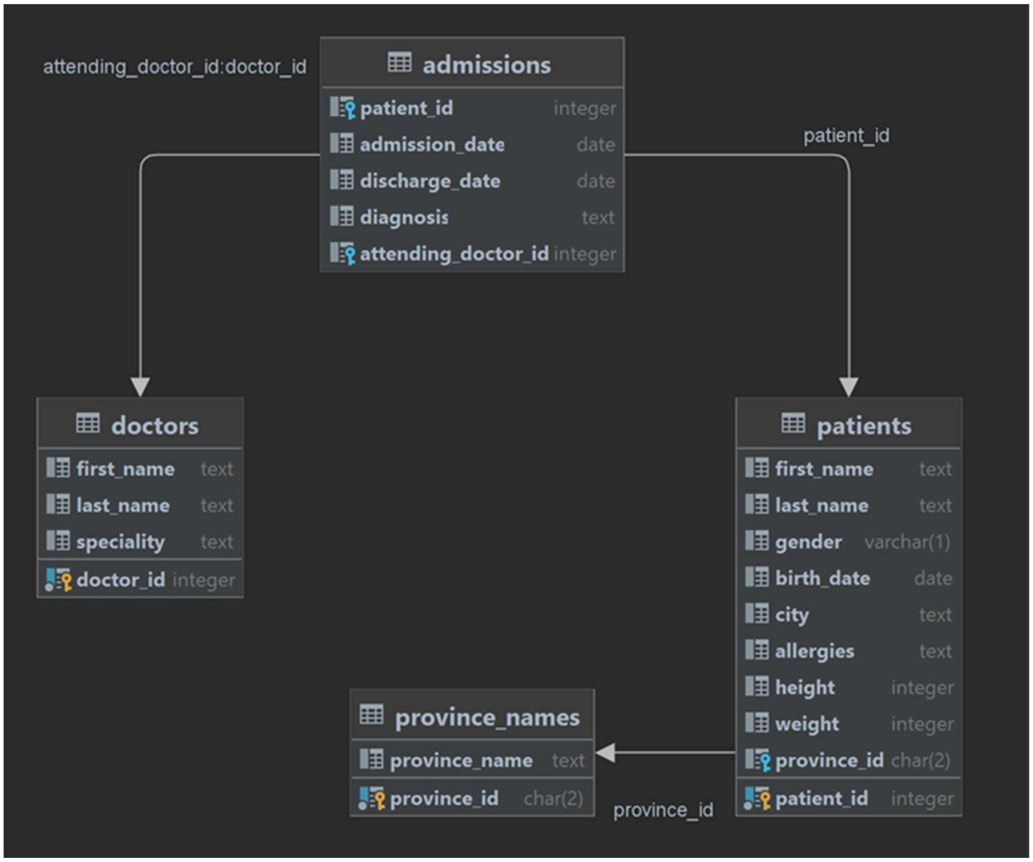
2. Tools & Technologies Used

- SQL (MySQL)
- Relational Database Management System (RDBMS)
- Database Schema Analysis
- Data Querying and Manipulation Techniques

3. Project Objectives

- Analyse patient demographic information.
- Investigate hospital admission patterns and trends.
- Examine diagnosis-related patient records.
- Explore doctor-patient relationships.
- Perform data cleaning and transformation operations.
- Generate statistical summaries using SQL.
- Apply advanced SQL techniques to solve real-world healthcare business problems.

4.database schema



5.Database Tables Used

1. Patients Table – Stores patient demographic and health-related information.
2. Admissions Table – Contains hospital admission and diagnosis records.
3. Doctors Table – Stores doctor details and medical specialties.
4. Province Names Table – Contains province information for geographical analysis.

6. SQL Concepts Demonstrated

Concept	Queries
SELECT & WHERE	1-5
String Functions	6,18,20,28
Aggregations	8,11,13,21,24
GROUP BY & HAVING	17,23,24,29
Joins	7,25,34
Date Functions	8,16,27,31
CASE Statements	33
UPDATE Statement	5
Advanced Analysis	23,29,31,34

7.Sample Queries

Query 1: Male Patients

Purpose: Retrieve the first name, last name, and gender of all male patients from the patients' table.

SQL query:

select first_name, last_name, gender

from patients

where gender='M';

Output:

first_name	last_name	gender
Donald	Waterfield	M
Mickey	Baasha	M
Jiji	Sharma	M
Blair	Diaz	M
Charles	Wolfe	M
Thomas	ONeill	M
Sonny	Beckett	M
Cedric	Coltrane	M
Hank	Spencer	M
Rick	Bennett	M
Woody	Bashir	M
Tom	Halliwell	M

Result:
Displays a list of all male patients along with their names and gender information.

Query 2: Patients Without Allergies

Purpose:
Identify patients who do not have any recorded allergies.

SQL Query:

select first_name,last_name

from patients

where allergies is null;

Output:

first_name	last_name	allergies
Donald	Waterfield	NULL
Blair	Diaz	NULL
Thomas	ONeill	NULL
Sonny	Beckett	NULL
Cedric	Coltrane	NULL
Hank	Spencer	NULL
Sara	di Marco	NULL
Amy	Leela	NULL
Rachel	Winterbourne	NULL
John	West	NULL
Jon	Doggett	NULL
Angel	Edwards	NULL

Result:
Returns the names of patients whose allergy information is missing or not recorded.

Query 3: Patient and Province Information

Purpose:
Display patient details along with their corresponding province name.

SQL Query:

```
select p.first_name,p.last_name,n.province_name
 from patients p join province_names n
 on p.province_id=n.province_id;
```

Output:

Result Grid | Filter Rows:

first_name	last_name	province_name
Donald	Waterfield	Ontario
Mickey	Baasha	Ontario
Jiji	Sharma	Ontario
Blair	Diaz	Ontario
Charles	Wolfe	Ontario
Sue	Falcon	Ontario
Thomas	ONeill	Ontario
Sonny	Beckett	Nova Scotia
Sister	Spitzer	Ontario
Cedric	Coltrane	Ontario
Hank	Spencer	Ontario
Sara	di Marco	Ontario

Result 7 ×

Result:
Provides a combined view of patient information and their province of residence.

Query 4: Count Patients Born in 2010

Purpose:
Determine how many patients were born in the year 2010.

SQL Query:

```
select count (*)
 from patients
 where year(birth_date)=2010;
```

Output:

Result Grid | Filter Rows:

count(*)
55

Result:
Returns the total number of patients born in 2010.

Query 5: Tallest Patient

Purpose:
Find the patient with the maximum height.

SQL Query:

```
SELECT first_name,last_name,height
FROM patients
WHERE height =(SELECT MAX(height)
   FROM patients);
```

Output:

Result Grid | Filter Rows:

first_name	last_name	height
Sam	Haruko	226

Result:
Displays the tallest patient and their height

Query 6: Total Admissions

Purpose:
Calculate the total number of hospital admissions recorded in the database.

SQL Query:

```
SELECT COUNT(*) AS total_admissions
FROM admissions;
```

Result Grid | Filter Rows:

Total_admissions
5067

Result:
Returns the total count of admission records.

Query 7: Cities with Highest Patient Count

Purpose:
Analyze patient distribution across different cities.

SQL Query:

```
select city,count(distinct(patient_id))

as number_of_patients from patients

group by city

order by number_of_patients desc,city;
```

Output:

city	number_of_patients
Hamilton	1938
Toronto	317
Burlington	276
Brantford	147
Ancaster	117
Stoney Creek	107
Cambridge	79
Dundas	79
Milton	65
Paris	58
Grimsby	55
Timmins	53

Result:
Shows cities ranked by the number of patients residing in each city.

Query 8: Most Common Allergies

Purpose:
Identify the most frequently reported allergies among patients.

SQL Query:

```
select allergies,count(patient_id)

number_of_patients

from patients

where allergies <> 'NKA'

group by allergies

order by count(patient_id) desc;
```

Output:

Result Grid | Filter Rows:

allergies	number_of_patients
Penicillin	1082
Codeine	305
Sulfa	157
ASA	99
Sulfa Drugs	71
Peanuts	52
Iodine	48
Tylenol	42
Bee Stings	40
Valporic Acid	38
Tetracycline	34
Wheat	33

Result:
Ranks allergies from most common to least common

Query 9: Obesity Classification

Purpose:
Classify patients as obese or non-obese based on BMI calculations.

SQL Query:

select patient_id,weight,height,

case

when weight/power(height/100,2)>= 30

then 1

else 0

end as is_obese

from patients;

Output:

patient_id	weight	height	is_obese
1	65	156	0
2	76	185	0
3	106	194	0
4	104	191	0
5	10	47	1
6	5	43	0
7	117	180	1
8	105	174	1
9	95	173	1
10	61	157	0
11	74	158	0
12	46	145	0

Result:
Displays a binary obesity indicator (1 = Obese, 0 = Not Obese) for each patient.

Query 10: Epilepsy Patients Treated by Dr. Lisa

Purpose:
Identify patients diagnosed with Epilepsy who were treated by a doctor named Lisa and display the doctor's specialty.

SQL Query:

select p.patient_id,p.first_name,p.last_name,d.specialty

from patients p

join admissions a

on p.patient_id=a.patient_id

join doctors d

on a.doctor_id=d.doctor_id

where a.diagnosis='Epilepsy'

and d.first_name='Lisa';

Output:

Result Grid | Filter Rows: | Export:

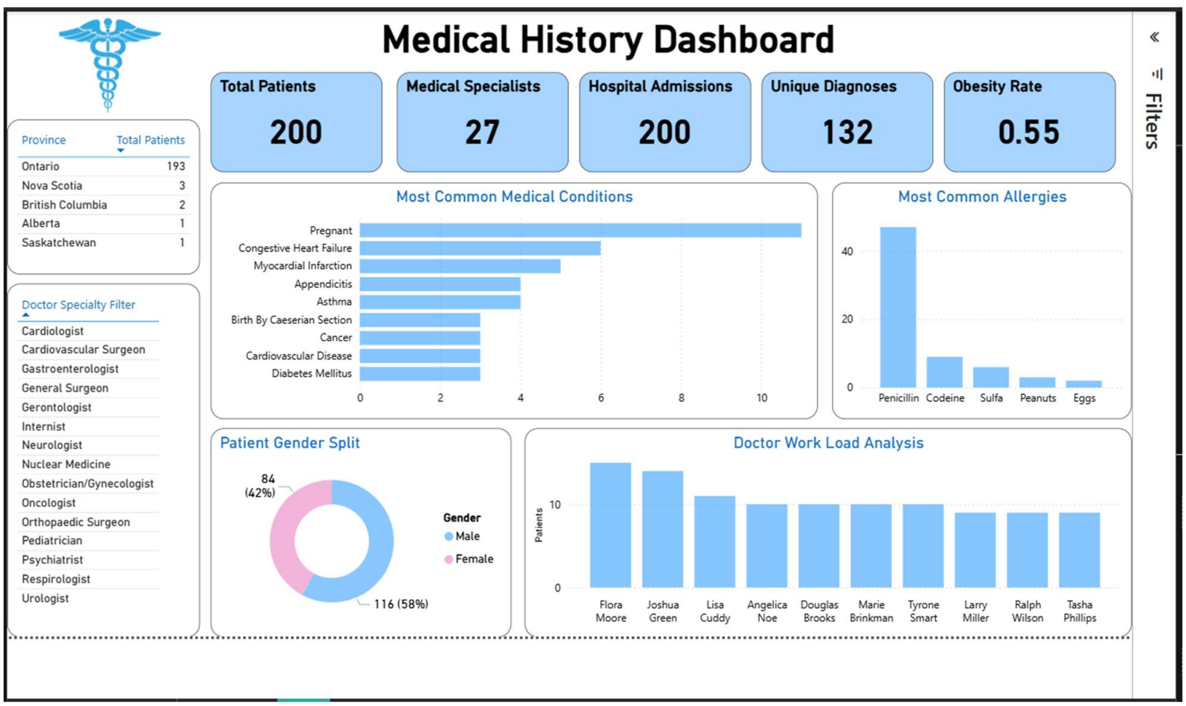
patient_id	first_name	last_name	specialty
468	Frank	Anderson	Obstetrician/Gynecologist
701	Precious	Ashton	Obstetrician/Gynecologist

Result:
Lists epilepsy patients along with the specialty of the attending doctor named Lisa.

8. Key Findings

- Retrieved patient demographic information.
- Analysed hospital admission trends.
- Identified repeated diagnoses and patient visits.
- Evaluated doctor specialties associated with specific diseases.
- Performed obesity classification using BMI calculations.
- Generated province-wise and city-wise patient statistics

9. Dashboard



10. Conclusion

The project provided practical experience working with a relational healthcare database. Through 34 business-oriented SQL queries, I strengthened my understanding of data extraction, manipulation, aggregation, and analytical problem-solving using SQL. The project demonstrates the ability to convert raw database records into meaningful insights for decision-making.